Eric Taylor

Oxford, UK • eric.taylor@worc.ox.ac.uk • LinkedIn

EDUCATION

University of Oxford, Worcester College – *DPhil in AI for the Environment* 2026–Present
Intelligent Earth CDT; UKRI-funded

Imperial College London – *MSc Environmental Data Science and Machine Learning* 2025–2026
Distinction

University of Cambridge, Pembroke College – *BA (Hons) Geography* 2022–2025
First-Class Honours with Distinction; ranked 1st in Physical Geography

RESEARCH EXPERIENCE & TECHNICAL PROJECTS

Master's Thesis, Imperial College London and Moody's Analytics Apr–Aug 2026

Domain-informed deep learning for downscaling tropical cyclone waves to hazard-relevant spatial scales.

Supervisors: George Pouliasis, Professor Matthew Piggot

- Developed a PyTorch framework for downscaling tropical-cyclone wave fields from 8 km to 1 km resolution, comparing deterministic U-Nets and Wasserstein GANs.
- Designed bathymetry-relative directional features, residual architectures and physics-based loss functions.
- Evaluated model generalisation across unseen storms, coastal geometries and extreme wave conditions, focusing on nearshore performance.
- Achieved an 85.8% reduction in nearshore mean absolute error relative to bilinear interpolation.

Undergraduate Dissertation, University of Cambridge 2024–2025

Remote sensing early supraglacial debris evolution on a mountain glacier using an energy balance model, thermal imagery, and climate reanalysis data.

Supervisor: Dr. Rebecca Dell

- Developed an energy-balance inversion framework to estimate supraglacial debris thickness from thermal satellite imagery and climate reanalysis data.
- Quantified spatial and temporal debris evolution and analysed physical controls on debris-cover development.
- Awarded 91%; William Vaughan Lewis Prize; Highly Commended for RGS best UK undergraduate dissertation.
- Presented research at the 2025 International Glaciological Society British Branch meeting, Oxford.

Personal Technical Project – Sea-Ice Concentration Estimation Jan–Mar 2026

- Built and packaged a PyTorch U-Net for estimating sea-ice concentration from dual-polarisation SAR imagery (HH/HV), implementing patch-based training, masked regression losses and reproducible evaluation.

Research Assistant, University of Cambridge (Prof. Mike Hulme) Aug 2025

- Analysed large climate datasets in Python to investigate trends in UK heatwaves and cold waves, producing publication-quality figures for forthcoming research.

APPLIED EXPERIENCE

Stratospheric Platforms Limited – *Project Intern* Jul–Aug 2024

- Developed MATLAB spatial-modelling algorithms for 5G coverage and analysed UK population data to assess coverage requirements and performance trade-offs.

AWARDS

Philip Lake Prize – highest performance in Physical Geography, University of Cambridge 2025
William Vaughan Lewis Prize – First-Class dissertation (top 15% of marks) 2025
Pembroke College prizes and scholarships for First-Class exam results 2023–2025

LEADERSHIP & COMMUNICATION

Worcester College MCR – *Treasurer* 2026–Present
Cambridge University Geographical Society – *Treasurer (2023–2024) and President (2024–2025)*

- Led a 12-person committee, organised academic talks and social events, modernised the society website, and introduced new accounting systems.

Content Creator (YouTube) 2019–2023

- Built a Minecraft tutorial channel to 40,000 subscribers and 8M+ views, with a Discord server of 5,000 members.